8. Beadandó

1 Szekvent kalkulus (2 pont)

Bizonyítsa a következő szekventet C-kalkulusban!

$$\exists x R(x) \wedge \neg P(y), \forall x(P(x) \wedge Q(x,y)) \longrightarrow \exists x R(x) \wedge \exists x Q(x,y)$$

1.0.1 Levezetési szabályok

$(\to \supset)$ $\dfrac{X, \Gamma \to \Delta, Y}{\Gamma \to \Delta, (X \supset Y)}$ $\qquad$ $(\supset \to)$ $\dfrac{\Gamma \to \Delta, X \qquad Y, \Gamma \to \Delta}{(X \supset Y), \Gamma \to \Delta}$

$(\to \wedge)$ $\dfrac{\Gamma \to \Delta, X \qquad \Gamma \to \Delta, Y}{\Gamma \to \Delta, (X \wedge Y)}$ $\qquad$ $(\wedge \to)$ $\dfrac{X, Y, \Gamma \to \Delta}{(X \wedge Y), \Gamma \to \Delta}$

$(\to \vee)$ $\dfrac{\Gamma \to \Delta, X, Y}{\Gamma \to \Delta, (X \vee Y)}$ $\qquad$ $(\vee \to)$ $\dfrac{X, \Gamma \to \Delta \qquad Y, \Gamma \to \Delta}{(X \vee Y), \Gamma \to \Delta}$

$(\to \neg)$ $\dfrac{X, \Gamma \to \Delta}{\Gamma \to \Delta, \neg X}$ $\qquad$ $(\neg \to)$ $\dfrac{\Gamma \to \Delta, X}{\neg X, \Gamma \to \Delta}$

$(\forall \to)$ $\dfrac{[A(x \parallel t)], \forall x A, \Gamma \to \Delta}{\forall x A, \Gamma \to \Delta}$ $\qquad$ $(\to \forall)$ $\dfrac{\Gamma \to \Delta, A}{\Gamma \to \Delta, \forall x A}$ $(x \notin Par(\Gamma, \Delta))$

$(\exists \to)$ $\dfrac{A, \Gamma \to \Delta}{\exists x A, \Gamma \to \Delta}$ $(x \notin Par(\Gamma, \Delta))$ $\qquad$ $(\to \exists)$ $\dfrac{\Gamma \to \Delta, [A(x \parallel t)], \exists x A}{\Gamma \to \Delta, \exists x A}$